

# Optics Lens Assignment 1

## Instructions

**Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.**

### A. General:

1. This question paper has 15 questions
2. Additional sheets for rough work shall not be provided
3. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and any other electronic gadgets will not be allowed
4. Do not tamper / mutilate the ORS

### B. Filling the ORS:

5. Fill in your Roll No., Name and Signature with a pen in appropriate boxes. Do not write these anywhere else
6. Darken the appropriate bubbles below your ID

### C. Question paper format and Marking scheme:

7. The question paper is divided into 1 parts; Physics . Each part consists of two sections.
8. In Section A, each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct. For every correct choice marked, 4 marks shall be awarded, and for every wrong choice marked 0 mark will be deducted.
9. In Section B, the answer to each question is a numerical value. For every correct answer, 4 marks shall be awarded, and for every wrong answered 0 mark will be deducted.

## Physics

### Section A

Questions 1-15 contains questions with 4 options (1), (2), (3) and (4), of which only one is correct. Mark the most appropriate choice.

1. A plano-convex lens ( $f = 20\text{ cm}$ ) is silvered at plane surface. Now focal length will be
2. The focal length of a convex lens of refractive index 1.5 is  $f$  when it is placed in air. when it is immersed in a liquid it behaves as a converging lens of focal length becomes  $xf$  ( $x > 1$ ). The refractive index of the liquid

3. For a given convex lens the  $\frac{1}{u}, \frac{1}{v}$  graph for real object (u) and virtual image (v) is

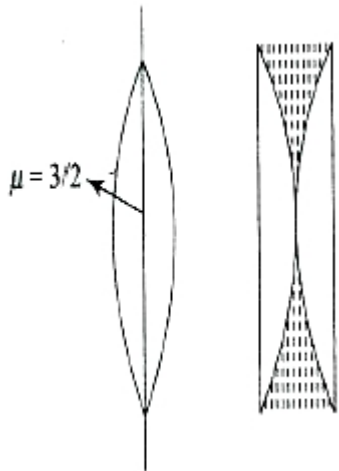
4. Arrange the following glass convex lenses in increasing order of their focal lengths\_\_\_\_\_

- a)  $R_1 = 12 \text{ cm}, R_2 = 6 \text{ cm}$
- b)  $R_1 = 10 \text{ cm}, R_2 = 5 \text{ cm}$
- c)  $R_1 = 15 \text{ cm}, R_2 = 5 \text{ cm}$
- d)  $R_1 = 15 \text{ cm}, R_2 = 10 \text{ cm}$

5. An object is placed at 20cm from a convex mirror of focal length 10 cm. The image formed by a mirror is\_\_\_\_\_

6. With a concave lens, an object is placed at a distance  $x_1$  from the principal focus, on the principal axis. The image is formed at a distance  $x_2$  from the principal focus. The focal length of the mirror is\_\_\_\_\_

7. A thin equiconvex lens ( $\mu = 3/2$ ) of focal length 10 cm is cut and separated and a material of refractive index 3 is filled between them. What is the focal length of the combination?



8. You are given two identical plano-convex lenses. When you place an object 20 cm to the left of a single plano-convex lens, the image appears 40 cm to the right of the lens. You then arrange the two plano-convex lenses back to back to form a double convex lens. If the object is 20 cm to the left of this new lens, what is the approximate location of the image?



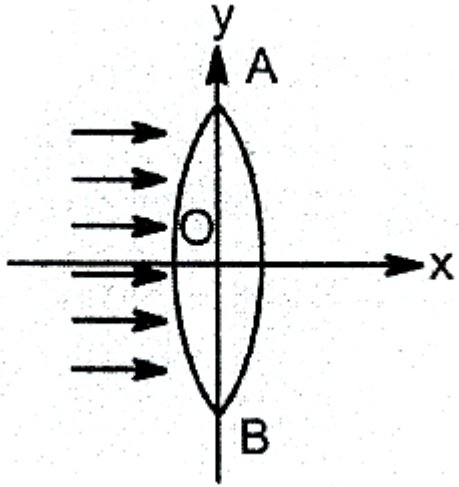
Plano-convex



Double convex

9. The power of a lens having refractive index 1.25 is  $+3$  diopters. When placed in a liquid its power is  $-2$  diopters. The refractive index of the liquid is :

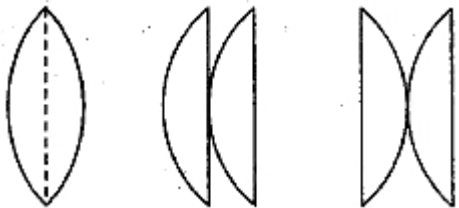
10. Monochromatic light rays parallel to  $x$  - axis strike a convex lens  $AB$  . If the lens oscillates such that  $AB$  tilts upto a small angle  $\theta$  (in radian) on either side of  $y$  - axis, then amplitude of oscillation of image will ( $f$  = focal length of the lens) :



11. A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of the combination is

12. Focal length of convex lens in air is  $16\text{cm}$  ( $\mu_{\text{glass}} = 1.5$ ). Now the lens is submerged in liquid of refractive index 1.42 . Find the ratio of focal length in medium to focal length in air has closest value

13. Two similar plano-convex lenses are combined together in three different ways as shown in the figure. The ratio of focal lengths in three cases will be



14. A convex lens of focal length,  $f$  is placed somewhere in between an object and a screen. The distance between the object and the screen is  $x$ . If the numerical value of magnification produced by a lens is  $m$ , focal length of a lens is

15. A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M. When pin is placed at A, such that  $OA = 18\text{ cm}$ , its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index  $\mu_l$  is put between the lens and the mirror, the pin has to be moved to  $A'$ , such that  $OA' = 27\text{ cm}$ , to get its inverted real image at  $A'$  itself. The value of  $\mu_l$  will be:

